125101 - Analytical chemistry 1 for engineers

Tutorial 6 – Acids /bases and Buffer solutions

T6-1. Calculate the pH of the following solutions:

a. CH_3COOH solution ($K_a=1.8\times10^{-5}$) with a concentration of 0.010 mol L^{-1};

b. $Cl_2CHCOOH$ solution ($K_a=10^{-1.26}$) with a concentration of 0.20 mol L^{-1}.

T6-2. Find the fraction of dissociation α of HAc ($K_a=1.75\times10^{-5}$) at the following pHs:

pH	α
2.00	
2.50	
3.50	
4.00	
4.76	
5.76	
6.00	
7.00	
8.00	

T6-3. The Kjeldahl method is an important method, in which we measure the amount of nitrogen in proteins, milk, cereals, etc.

In an experiment, 1.8 grams of ammonium decavanadate [$(NH_4)_6V_{10}O_{28}*6H_2O$] (1173.7 g/mol) were placed in a Kjeldahl container and cooked with an excess of NaOH to obtain ammonia. The ammonia was distilled and collected in a trapping flask containing 50 mL of 0.1M H_2SO_4. By the end of the reaction, the solution was titrated with 28 mL of 0.05 M NaOH.

What is the weight percentage of nitrogen in the sample? Compare the obtained value with the theoretical value from the formula weight.

$$(NH_4)_6V_{10}O_{28} * 6H_2O + 6NaOH \xrightarrow{\Delta} 6NH_3 + Na_6V_{10}O_{28} + 6H_2O$$

$$H_2SO_4\ (total) + 2NH_3 \rightarrow SO_4^{-2} + 2NH_4^+$$

T6-4. What is the pH of a buffer after adding 15.0 mL of 1.50 M HCl to 1.00L solution containing 12.43gr of "tris"(hydroxymethyl-aminomethane, B) (121.135 gr/mole) and 4.67gr of its conjugate acid BH^+ (157.596 gr/mol)? The tris pKa is 8.072.